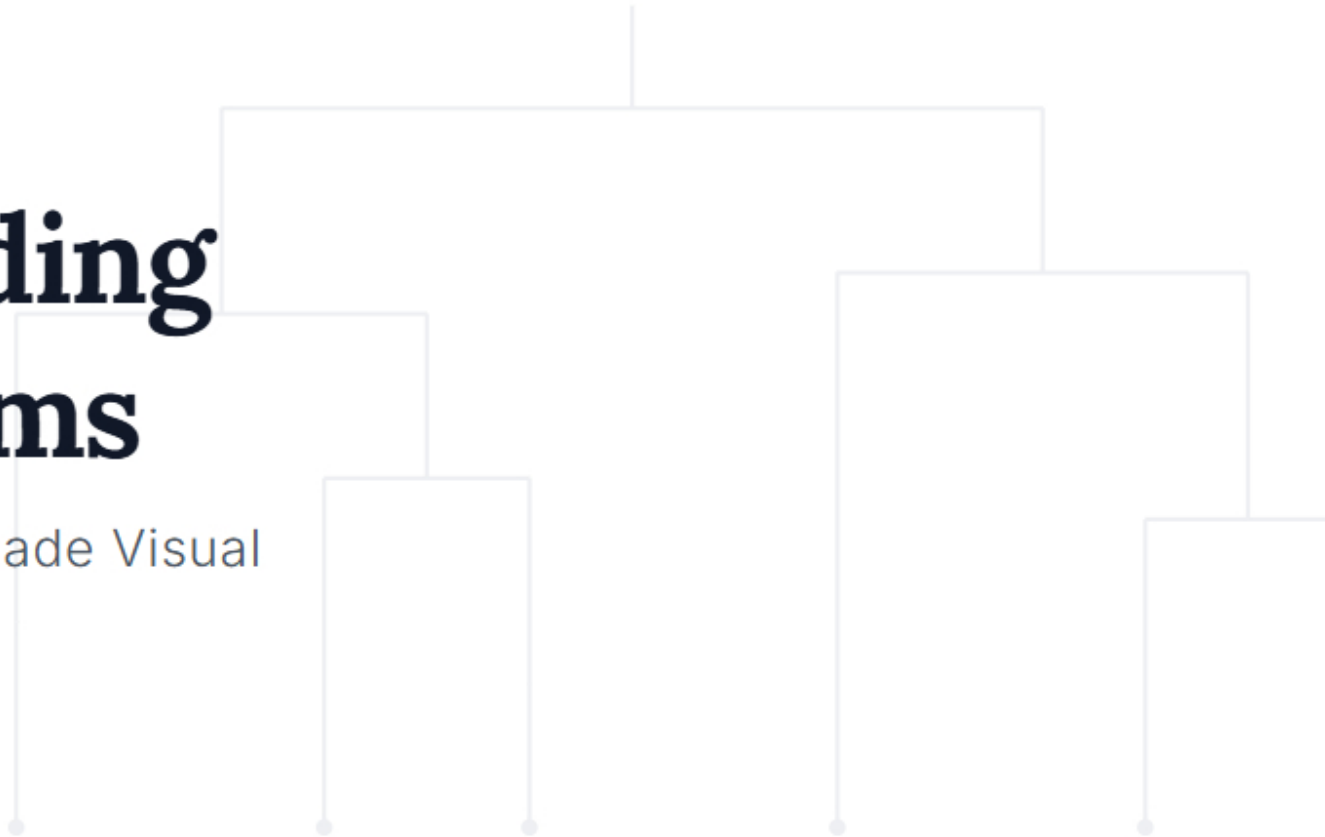


Understanding Dendrograms

Hierarchical Clustering Made Visual

PRESENTED BY

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What Is a Dendrogram?

A tree-like diagram that records the sequences of merges or splits in a hierarchical clustering analysis.

CORE APPLICATIONS



Biology & Phylogenetics

Visualizing evolutionary relationships and genetic similarities between species.



Data Science & Statistics

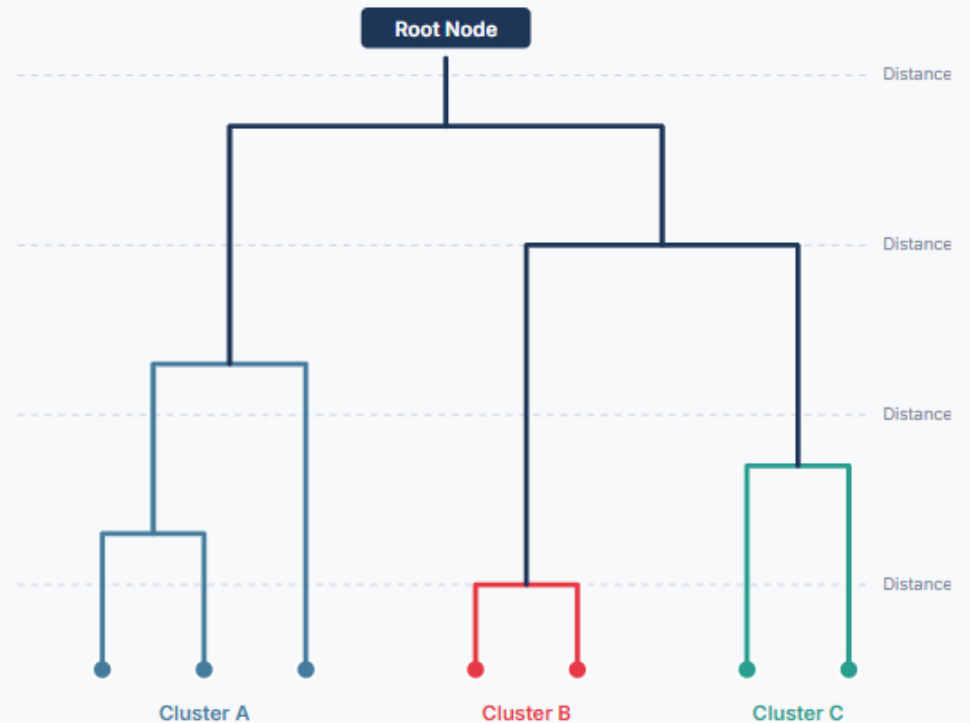
Understanding data structure before choosing the number of clusters (K).



Machine Learning

Feature grouping and hierarchical classification tasks.

VISUAL STRUCTURE



What a Dendrogram Shows

Every element in the diagram corresponds to a specific structural property of your dataset's hierarchy.



Leaves (Bottom Nodes)

Represent individual data points or objects at the start of the process (Distance = 0).



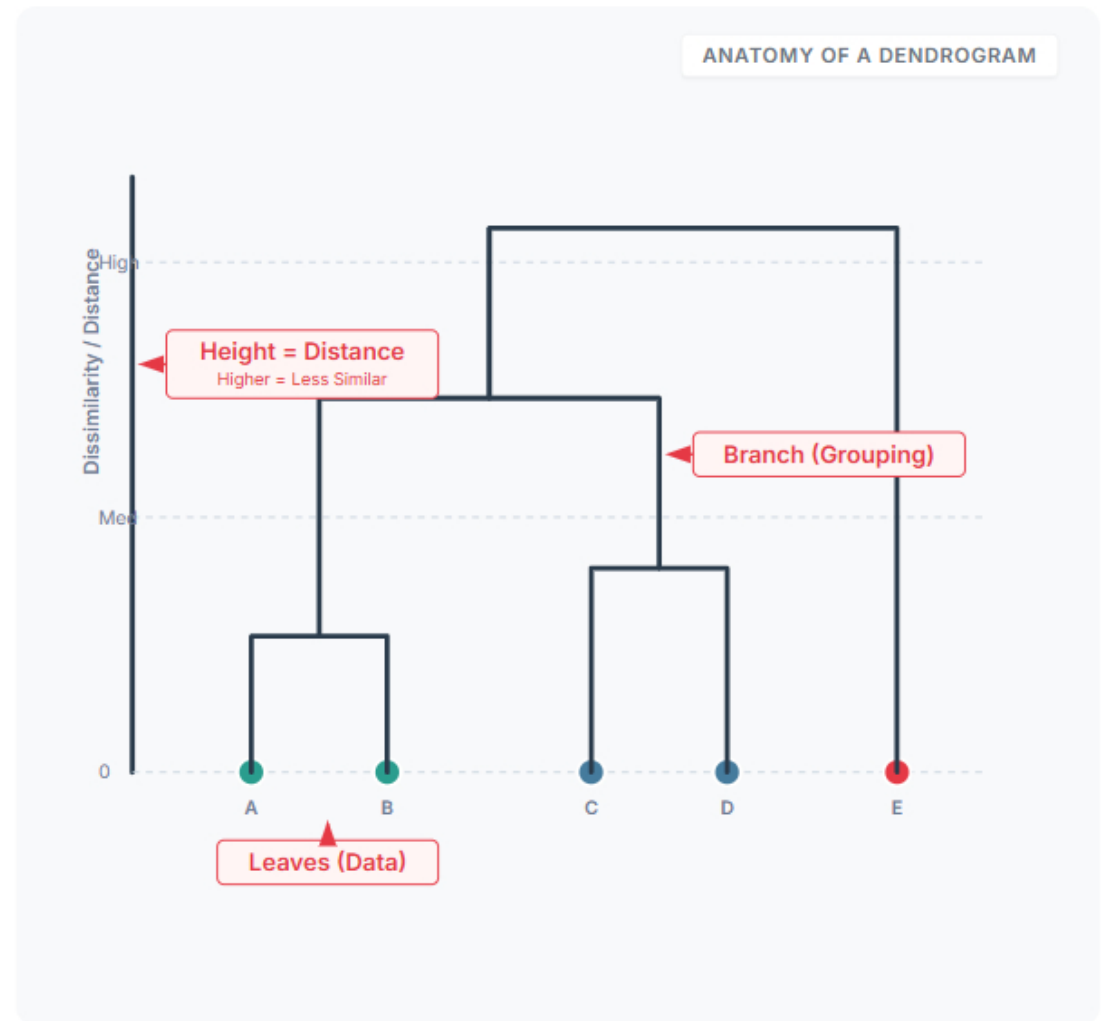
Branches & Clades

U-shaped lines connecting clusters. Horizontal lines indicate which clusters are merging.



Height of Merges

Represents the distance or dissimilarity. Higher merges = Less similar clusters.



Choosing Clusters by Cutting

To determine the final grouping, you "cut" the dendrogram horizontally. The number of vertical lines the cut intersects equals the number of clusters (k).

INTERPRETATION RULE

The height of the cut represents the threshold of dissimilarity. Groups merged below this line are considered single clusters.

High Cut

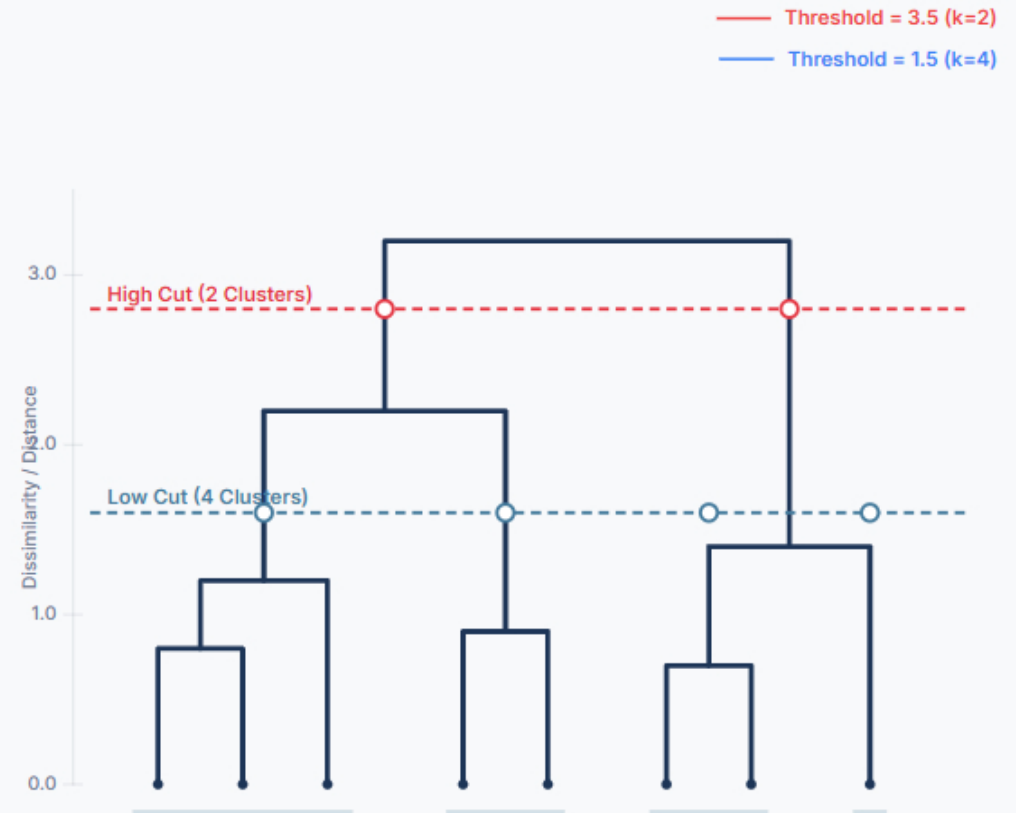
Fewer Clusters (Small k)

Captures broad, general categories. High dissimilarity allowed within groups.

Low Cut

More Clusters (Large k)

Captures specific, cohesive sub-groups. Low dissimilarity within groups.

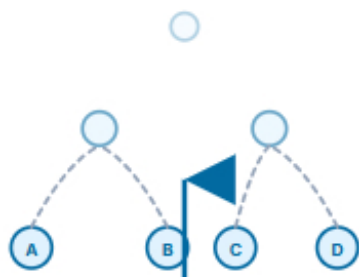


How Dendrograms Are Created

Two opposing approaches to building the hierarchy

Agglomerative

BOTTOM-UP APPROACH

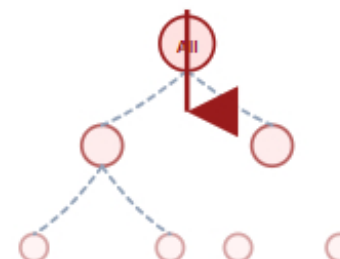


- **Start:** Treat every data point as its own individual cluster.
- **Process:** Iteratively merge the two "closest" clusters based on a distance metric.
- **Verdict:** Most common approach. Computational complexity is generally $O(n^3)$.

VS

Divisive

TOP-DOWN APPROACH



- **Start:** Begin with all data points contained in a single large cluster.
- **Process:** Recursively split the cluster into two most dissimilar groups.
- **Verdict:** Less common. Computationally expensive ($O(2^n)$) but can be more accurate for top-levels.

Advantages of Dendrograms

Why hierarchical clustering is a powerful analytical tool



Visual Clarity

Provides an intuitive map of relationships. Users can easily see the full hierarchical structure at a single glance.



No Predefined Clusters

Unlike K-Means, you do not need to define the number of clusters (\$k\$) beforehand. Explore structure freely.



Deep Insight

Reveals nested sub-structures and identifies outliers that flat clustering algorithms often miss.